Observer Patch Holography as a Strange-Loop Self-Simulation
The Claim Lattice and Its Closure Status

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Observer-Patch Holography (OPH) begins with one radical subtraction. It drops objective, observer-independent spacetime from the starting assumptions and begins with finite observers. Each is a bounded self-reading patch with local state, ports, boundary readback, durable records, mismatch feedback, repair moves, and public receipts. No patch contains the world. Reality is the stable public fixed point that survives when all local views are made mutually consistent,

$$T(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

The strange-loop principles [C17]

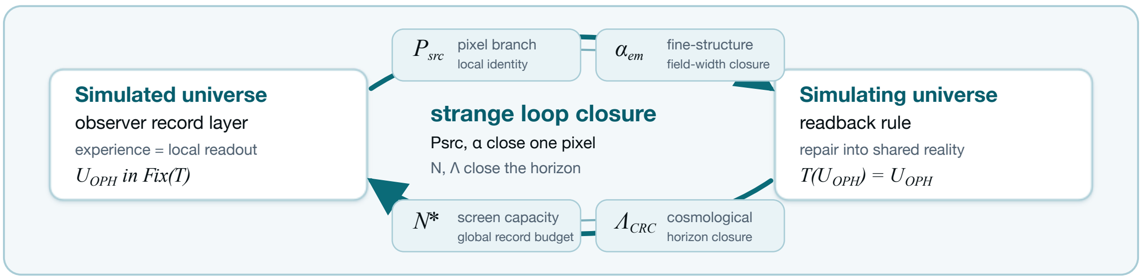
Every claim in this document is deduced from six declared strange-loop principles, condensed here from the canonical statement in the program's claim registry [C17]. The mathematical axiom sets inside the papers (the five-axiom recovered core, the patch-algebra axioms) are a different kind of object; the principles are the physical frame those theorem packages are read through. **SL-0 (Strange loop):** reality is the fixed point of a self-reading computation; the simulating structure and the simulated world are one closed system, with no external machine and no external clock. **SL-1 (Substrate chart):** the observer-facing substrate is charted by the two-sphere with the screen-microphysics architecture (patches, records, overlap comparison, repair); 3+1-dimensional observer experience is a design constraint of the substrate under SL-0. **SL-2 (Detuning law):** the pixel ratio sits off the golden-ratio balance point by the Gaussian-normalized width of one electromagnetic observation, $P = \varphi + \sqrt{\pi}\,\alpha$; a screen at exact balance carries no events. **SL-3 (Constant identity):** under self-simulation the fine-structure constant of the simulated world and the substrate pixel readout are one quantity; the theory computes P from its closure equation and the measured α is the test that value faces. **SL-4 (Capacity identity):** the total record capacity N is one number shared by substrate and world; the theory computes N from $N = F(N)$ and the Λ readout is the test that value faces. **SL-5 (SI bridge):** one clock anchor, the cesium transition, connects substrate units to laboratory units.

The theory layer takes no measured number: zero dials, zero quantitative inputs in every defining equation. Measured values enter at the test layer as comparisons, and at the working layer as located values inside lanes whose closure terms are open: the working pixel $P = 1.630968209403959\ldots$ located from CODATA $\alpha^{-1} = 137.035999177(21)$, and the working capacity $N = 3.31 \times 10^{122}$ located from the Λ readout. Every working borrow is tied to a named generator and retires when that generator lands.

Douglas Adams proposed that the Earth is a computer running a very long calculation, and he was joking. OPH makes the adjacent claim with the joke removed: the universe is a computation checking its own records, and the output is on public display. Take nothing except finitely many observers that keep records and argue until their notes agree, model the argument as a fixed-point computation on a holographic screen, and solve. Out falls a universe whose events obey Born-rule probabilities on a central record algebra [O1,O3], whose observers see $3+1$-dimensional Lorentzian spacetime running Einstein's equations [O4,C8], whose horizon closes at the de Sitter capacity $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$ so the sky reads as an accelerating expansion with a small cosmological constant [O1,O4], whose particle catalogue is the Standard Model quotient with three colors, three generations, and one Higgs doublet [O4], and whose galaxies rotate too fast for their visible mass in the pattern the repair-stress continuation prints on the empirically fitted MLS response (Section 5), SPARC χ^2 per point 1.017, so its residents budget for dark matter and dark energy and both line items read as bookkeeping on that fitted branch [O7,C9]. The punchline is the address. That universe resembles this one down to m_W, m_Z, m_t, and m_H on the source and conditional branches distinguished below; Newton's constant enters as an input through the SL-5 bridge. The historical charged-lepton continuation supplies an additional, compare-only numerical match. The calculation was not given the measured weak-boson masses: on the strict source-audit branch it returns $m_W = 80.330\,\mathrm{GeV}$ and $m_Z = 91.119\,\mathrm{GeV}$, within 4.9×10^{-4} and 7.6×10^{-4} of measurement, pulls of $\approx 2.9\sigma$ and $\approx 35\sigma$ in measurement sigma, carried as closure tension CL-5. The full answer runs to rather more decimals than 42.

The claim in plain words

The universe computes itself. Two numbers set the whole machine: a pixel ratio and a memory size, and neither is chosen by anyone. Each is the unique solution of an equation the universe must satisfy to contain its own simulation, and the uniqueness is proven, by computer-verified interval arithmetic, on the entire physical range. Run the machine and out come the shape of spacetime, the particle families, quantum probability, gravity, and numerical values that track the measured world to between a tenth of a percent and seven percent, with nothing anywhere to tune. What remains between this and exact agreement is a short public list: one counting theorem, one hadronic computation with a frozen pass/fail line, one electroweak repair. The theory cannot be adjusted while it waits, so every remaining item is a test it can fail. Theories built this way are either approximately right or briefly alive.



The case for identifying that output with our universe is carried by two maintained repository artifacts rather than by a probability estimate: the compression scorecard, which counts estimated settings against independently landed basins, and the closure ledger, which records the residuals the principle set requires to vanish. The reuse matters because the same normal form, compact-gauge branch, pixel endpoint, and edge-entropy normalization feed multiple sectors. Section 7 states the current standing of both artifacts.

This document presents twenty landmark solution claims and places eighteen quantitative outputs beside measurement, each with its relative offset and its pull in measurement sigma. Their evidence classes range from internal theorems to explicitly conditional comparisons and are labeled row by row. No probability estimate appears anywhere in this claim lattice.

Throughout, *derived* denotes an output of a declared source map whose inputs and branch rules are fixed independently of the comparison target. Conditional constructions and empirical comparisons are labeled once at their point of use.

20 landmark solutions, 18 measured comparisons

The auditable input statement applies to the rows labeled derived or source-audit: after the six strange-loop principles (SL-0–SL-5), the two estimated settings (P from measured α via SL-3; N from measured Λ via SL-4), and the declared discrete receipt branch are frozen, those rows introduce *zero additional target-fitted continuous parameters.* The closure coordinates are solutions,

$$P_\star = \Gamma_P(P_\star), \qquad N_\star = \Gamma_N(N_\star),$$

and the integers $(N_c, N_g, \beta_{\mathrm{EW}}) = (3, 3, 4)$ are branch outputs. Each source-derived sector reads the same normal form without a separate adjustable decimal. Conditional, external-input, and compare-only rows are kept outside that statement.

The positive claim spine

The compact argument follows six steps:

$$\begin{gathered}
\text{finite observer patches} \longrightarrow T(\mathfrak{U}) = \mathfrak{U} \longrightarrow \text{central record algebra}, \\
S^2 \text{ cap geometry} \longrightarrow \mathrm{SO}^+(3,1) \longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \\
\text{source closure} \longrightarrow (P_{\mathrm{fwd}}, N_\star), \\
\text{compact-gauge reconstruction} \longrightarrow \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \\
(P_{\mathrm{fwd}}, \alpha_U, 3, 3, 4) \longrightarrow \frac{v}{E_\star}, \qquad \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}} \longrightarrow G_{\mathrm{geom}}.
\end{gathered}$$

The companion papers carry the full theorems and executable receipts.

The selection chain and the uniqueness endgame

The claim lattice has one logical shape, stated in full in the consistency stack [C15]: ten consistency requirements C1–C10 quotient the space of self-consistent worlds. Self-reading forces records and repair (C1); overlap consensus forces the public quotient, and on the S^2 chart forces signature, dimension, and the Lorentz group (C2); modular self-consistency forces internal time and thermality (C3); transportable-charge consistency forces the compact gauge menu and, with the declared matter package, the Standard Model quotient (C4); entropic consistency forces the Einstein branch and leaves one global number (C5); record existence forces the detuning law and leaves one local number (C6). After C1–C6 the surviving freedom is the two-real-parameter family (P, N) together with the declared discrete selections the scorecard counts. C7 identifies the two parameters with measured readouts; C8–C9 are two equations on the family; C10 removes units. Three lemmas close the chain. **L1**: the pixel closure has exactly one fixed point per readout map on the declared physical domain $\alpha^{-1} \in [100, 200]$: existence and local uniqueness by interval Banach (the stage-2 contraction certificate [C1]), global at-most-one by certified $\sup |g'| < 1$ on all 256 subdivision pieces with zero exceptional set (the global uniqueness certificate) [C1]; **L2**: the same schema applies to N once the readback map F is constructed (CL-7; required properties given by the readback specification [C16]); **L3**: under L1+L2 the principle set admits at most one (P, N); for P this is now a theorem on the declared physical domain, and there is no landscape to relocate into: a blind closure landing outside its basin falsifies the formulation permanently. If the two equations close and the fixed points are unique, the principle set admits exactly one universe. Whether that universe is ours is the closure ledger reading zero: a theorem-shaped claim whose proof obligation is an experiment: armed, blind, and externally timestamped.

The two-part proof architecture

The claim decomposes into two theorems, and the program is the distance to each. **Part A (the loop forces its constants).** A self-simulating universe that contains its own simulation admits exactly one (P, N), and those values are the fine-structure and capacity readouts of the world inside it. Status: uniqueness is proven for P (exactly one fixed point per readout map on the physical domain, interval-certified [C1]); for N the coupling theorem G2-GAP-1 is proven modulo three declared premises: the seed ($N_0 = \pi$), the contraction, and the tick-projection identity are theorems, the algebraic layer is machine-checked in Lean, and the conditional fixed point is certified at $3.5321315434 \times 10^{122}$ with relative width 1.6×10^{-25}; the remaining core is one counting-theorem-shaped premise, the balance condition CP-1 [C16]. Identification with the measured readouts is the closure ledger: the α window is armed at 2.1×10^{-8} with the first-principles bracket already containing the required screening (CL-1), and the capacity comparison stands at 6.6% (CL-3). **Part B (the forced constants produce our universe).** Given (P, N), the observed universe follows. Status: the structural skeleton is theorem-grade on the declared branches: the Lorentz group, $3 + 1$ dimensions, the Standard Model quotient, three colors, three generations, one Higgs, and the structural zeros, with the named gates open (realized-branch nonemptiness, quantum pole receipts). The quantitative record stands between exact and 7%. Machine formalization covers the consensus core plus the coupling algebra (98 plus 13 sorry-free Lean theorems [C6]); extending it through the physics branches is Part B's open formalization program. The complete step-by-step spine with its gap register is the program's proof spine [C17]. **Completion criterion.** Part A closed by a landed identification plus Part B closed by discharged gates is the formal proof. This document carries the exact distance to each; the coincidence pricing below is supporting evidence, and the frozen experiments are the decisive instrument.

What the stack already delivers

Proven and machine-checked. The pixel closure has exactly one solution per readout map on the physical domain, by interval arithmetic. The tick-projection identity, the readback seed, and the bridge algebra are theorems; 111 Lean lemmas hold sorry-free. The SM coefficient triple and $\beta_{\mathrm{EW}} = 4$ are consistency-forced, their alternatives structurally excluded by a preregistered sweep. The conditional capacity theorem forces $N = 3.53 \times 10^{122}$ modulo one counting premise. **Hard problems, addressed structurally.** The measurement problem: Born probabilities and state update are the record algebra of consensus, proven on the declared branch. The cosmological constant problem: Λ is a finite capacity readout, $\Lambda \ell_\star^2 = 3\pi/N$, with no 10^{120} vacuum-energy fine-tuning anywhere in the theory, currently matching the capacity readout to 6.6%. The hierarchy problem: $v/E_\star \approx 2 \times 10^{-17}$ is a one-line transmutation law, matched to 2.3×10^{-4} with no supersymmetric machinery and no tuning. The dark sector: repair bookkeeping reproduces the radial-acceleration phenomenology on the fitted branch at 0.13 dex. Time: internal modular readout, with no external clock to postulate. The Yang–Mills gap: a finite repair mechanism, conditionally identified with the continuum gap. Each claim carries its branch conditions in Section 5. **The record.** Eleven quantities between exact and 7% from a chain with zero dials; the structural skeleton (Lorentz, $3+1$, the SM quotient, three colors, three generations, one Higgs) forced on declared branches; coincidence priced at one in thirty thousand under concessive assumptions; and every open gap enumerated in a public register with a frozen, timestamped decisive experiment.

Where this stands among theory-of-everything candidates

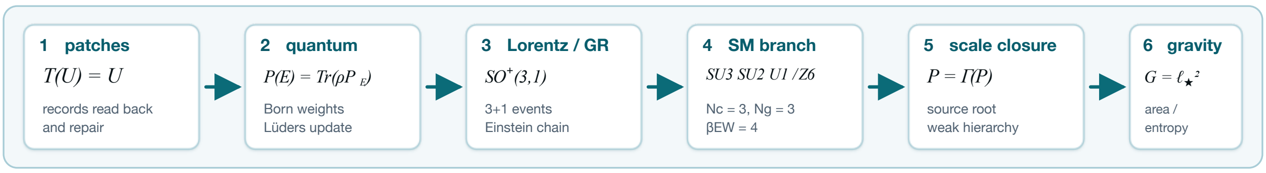
The comparison below is structural and factual; every OPH cell carries an artifact in this repository.

	SM + GR	**String / M-theory**	**Loop quantum gravity**	**OPH**
Adjustable parameters	about 26 dials, values unexplained	none in principle, then a landscape of $\sim 10^{500}$ vacua	scope excludes the constants	zero dials; both constants are proven-unique fixed points
Constants of nature	inputs	environment-selected on the landscape	outside scope	forced by closure: α to 3×10^{-4}, capacity behind Λ to 6.6%, exact closure armed
Uniqueness of the solution	n/a	no selection principle	n/a	exactly one per map on the physical domain, certified
Can it die tomorrow	no single kill test	relocates within the landscape	no sharp kill test	frozen, timestamped blind targets; executed failures kept public with permanent verdicts
Machine-checked core	no	no standard artifact	partial	111 sorry-free Lean theorems
Hard-problem record	measurement, Λ, hierarchy open	Λ anthropic; hierarchy via SUSY, strained	gravity sector only	measurement, Λ, hierarchy, dark sector, time: addressed structurally, branch conditions on record
Remaining distance	undefined	undefined	undefined	a finite public gap register with dated statuses

The practice of freezing pass/fail targets before computing payloads, and of keeping executed failures on a permanent public ledger, is imported from experimental science and has no counterpart in the standard theory-of-everything toolkit. A candidate that cannot be tuned, cannot relocate, and publishes the exact list of ways it can die is playing a different game from its competitors, and the ledger says how the game stands on any given day.

Five facts and one experiment

One. The theory carries zero dials and, at the theory layer, zero quantitative inputs: both constants are defined by closure equations, and nothing in the architecture can be adjusted to fit data. **Two.** The pixel closure has exactly one fixed point per readout map on the declared physical domain (interval arithmetic, outward rounding, local $L \leq 0.0724$, global $\sup |g'| < 1$ on every subdivision piece) [C1]. **Three.** With nothing to tune, the forced chain currently reproduces the fine structure constant to 3×10^{-4}, the W and Z masses to 4.9×10^{-4} and 7.6×10^{-4}, the hierarchy relation to 2.3×10^{-4}, the record capacity behind Λ to 6.6×10^{-2}, and the strong scale to about 10^{-2}, with every discrete selection counted in the scorecard. **Four.** The distance to exact closure is finite and enumerated: three open generators and one selector, and all four were driven forward on 2026-07-14. G3: a preregistered 96-entry electroweak revision sweep exhausted the certified-pixel menu and showed the SM coefficient triple admits no fixed point at all, so that selection is consistency-forced. G2: every declared-structure readback candidate was certified and excluded, and the unique surviving reading reproduces the electroweak bridge expression, naming CL-7 as one coupling theorem. G1: the payload harness exists, and the first-principles internal-mass bracket contains the required screening. The selector: twelve preregistered candidates evaluated, all excluded. Row CL-6 is closed. **Five.** The decisive test is armed. The blind hadronic target is frozen at precision 160, externally timestamped, and carries a closed-form pass tolerance and a payload-work-start rule, so neither side of the comparison can move after the fact. **The experiment.** A chain with no dials cannot be adjusted into agreement, so the record above is either the loop showing through or a coincidence, and the two readings separate at the frozen window.



Eighteen quantitative comparisons

Source endpoints and declared conditional comparisons, side by side with measurement. Evidence classes are tagged per row and stated in full in Sections 3 and 5.

Quantity	OPH	Reference	Offset	Pull (σ)
m_W	80.330 GeV	80.3692(133) GeV	-4.9×10^{-4} (a)	2.9
m_Z	91.119 GeV	91.1880(20) GeV	-7.6×10^{-4} (a)	≈ 35 †
m_t	172.63 GeV	172.60(30) GeV	$+2\times10^{-4}$ (b)	0.1 ‡
m_H	125.77 GeV	125.13(11) GeV	$+5.1\times10^{-3}$ (b)	5.8 †‡
m_H at measured m_t	125.72 GeV	125.13(11) GeV	$+4.7\times10^{-3}$ (b)	5.4 †‡
m_μ/m_e	206.7703160	206.76828	9.83 ppm (c)	≈ 440 †
m_τ/m_e	3477.4728	3477.365	30.94 ppm (c)	0.5
(m_e, m_μ, m_τ) centrals	0.5089/105.22/1769.5 MeV	0.5110/105.66/1776.9 MeV	-0.42% common (d)	$\gg 10^3$ †
$\Lambda^{(3)}_{\rm QCD}$	334.8 MeV	338(12) MeV	-0.9% (e)	0.3
m_N	929 MeV	938.9 MeV	-1.0% (e)	0.3
$G_{\rm SI}$	$6.6742999959\times10^{-11}$	$6.67430(15)\times10^{-11}$	identity (e)	input §
EW capacity $\to\Lambda$	2.668×10^{-122}	2.845×10^{-122}	6.6% gap in Λ units and in N units (e)[S12,C7]	CL-3
$\alpha_s(M_Z)$	0.11834	0.1179(9)	$+0.5\sigma$ (i)[C7]	0.5
Fixed $\mathbb{Z}_6$ RAR	0.13283 dex RMS	same-data optimum 0.13281	2×10^{-5} dex (e)[C7]	‡
BTFR slope	4	3.8457(858)	$+1.80\sigma$ (e)[S14]	1.8
Scalar tilt n_s	0.9660215	0.9649(42)	$+0.27\sigma$ (h)[S12]	0.3
$\Delta\alpha^{(5)}_{\rm had}(M_Z^2)$	0.02676(75)	0.02761(11)	1.1σ (f)	1.1
Cosmic birefringence	$\alpha_U/(2\pi)=0.37501^\circ$	$0.342^{+0.094}_{-0.091}{}^\circ$	$+0.35\sigma$ (j)[C7]	0.35

(a) strict source-audit branch, no measured mass input; (b) criticality branch, determined top; (c) compare-only balanced-carrier postdiction; (d) certified empirical-closure interval; (e) conditional, external-input, or retrospective branch; (f) declared-empirical compilation; (h) conditional analytic screen branch, with source, radial-lift, and official-likelihood gates unproved; (i) conditional source-running branch; (j) S1 retrospective coincidence, four trials declared. The pull column restates each offset in units of the reference measurement sigma; for conditional rows whose reference sigma is negligible the pull uses the propagated input uncertainty. †: pull exceeds 3σ, so the row is excluded as prediction at current precision; the electroweak rows are carried as closure tension CL-5 in the closure ledger [C17], and the interval row (d) remains a certified containment. ‡: calibration, meaning the row's construction consumed its comparison target (the D11 top/Higgs census; the same-data RAR optimum). §: SL-5 bridge input, the unit-bookkeeping identity on the selected scale bridge; G rides the bridge the way c and the SI second ride every physical theory. The same ledger gives the dependent cross-check $m_t/m_W = 2.1490$ versus 2.1476 and the S1 diagnostic $\ell_{\rm IR}=32$, only 0.2 χ^2 units from the low-ℓ optimum. Neither is counted as an independent prediction [C7].

1. The observer-patch fixed point

The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form [O1,O2]. Let T be the observer-overlap repair map and let $\mathfrak{U}_{\rm OPH}$ denote the selected normal form. The simulation identity is

$$T(\mathfrak{U}_{\rm OPH}) = \mathfrak{U}_{\rm OPH}.$$

The OPH simulation theorem establishes this identity from endogenous update, observer-readable records, schedule-independent overlap repair, selected-branch elimination, and implementation/clock closure [O2]. The universe is the stable record world returned by its own readback-and-repair computation.

Conditional closure theorem

Fix the strange-loop principles, branch rules, and quotient/readout maps. Let

$$\Gamma_P : I_P \to I_P, \qquad \Gamma_N : I_N \to I_N$$

be source-only contractions selected uniquely by those rules up to observer-invisible equivalence. Banach's theorem gives unique endpoints

$$P_\star = \Gamma_P(P_\star), \qquad N_\star = \Gamma_N(N_\star).$$

Every declared observable

$$Y_i = \pi_i(\mathrm{nf}_{\rm OPH}(P_\star, N_\star))$$

is therefore fixed by the selected branch and invariant under admissible implementation and schedule changes. Data comparison identifies the physical branch represented by these source outputs.

Status of the two hypotheses: on the P side the contraction hypothesis is discharged on an explicit interval by the interval-arithmetic certificate in [C1] (stage 2 of the basin-then-contract protocol); on the N side it is conditional on constructing the readback map F (CL-7), whose required properties (domain, monotonicity regime, growth bounds, and blindness under the dependency-cone audit) are specified in [C16].

The concrete carrier is a finite collection of local states, ports, central records, repair maps, and checkpoints. The reference benchmark replays all six overlap schedules and returns the same repaired normal form on the clean abelian branches [C5]. A separate Lean 4/Mathlib artifact certifies observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector in 98 sorry-free theorems [C6].

For observer patch i, the readout equations are

$$r_i = \mathcal{R}_i(\mathfrak{U}_{\rm OPH}), \qquad \mathfrak{U}_{\rm OPH} = T(\{r_i\}_{\rm compat}).$$

The first map gives the local record; the second repairs compatible local records into the public normal form. This is the observer-like self-reading structure specific to OPH: local records participate in the computation that returns the world those records describe.

Quantum event surface

On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E,

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E), \qquad \rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

These are the Born rule and Lüders update. On the associated two-wing Bell surface,

$$|S_{\mathrm{CHSH}}| \leq 2\sqrt{2}.$$

A finite 65,536-record run reproduces the event algebra numerically: committed-event weights sum to 1.0, record projectors have zero idempotency defect, and repeat reads are stable [O1,O3,C7].

2. Relativity, (3+1) events, and Einstein dynamics

The screen readout supplies an oriented conformal S^2. Its connected conformal group is the connected Lorentz group,

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

The corresponding observer-frame space is

$$H^3 \cong \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 6 - 3 = 3.$$

On the declared event branch, repaired records populate a four-dimensional event base of signature $(-{+}{+}{+})$, while H^3 is the fiber of unit timelike observer frames over each event [O4].

The 2π-normalized modular flow of screen caps supplies the observer-relative clock. Half-sided modular inclusions give positive null translations; null tomography assembles their directional charges into a conserved local T_{ab}. Fixed-cap generalized-entropy stationarity, the uniform small-ball limit, and the all-directions tensor upgrade then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle.$$

Every arrow in this chain has a named theorem and a machine-readable receipt type [O4,C8].

Earned simulation receipts at 64k, 128k, and 1M patches

Three canonical runs instantiate bounded self-reading patches and scale from 65,536 patches/1,024 observers through 131,072/32,000 to 1,048,576/64,000 [C7]. The 128k and 1M runs earn observer-local modular time, gauge-covariant 2π-KMS replay, a conformal H^3 chart, controlled H^3 response, integer $3+1$ observer-facing experience, mutual agreement, and an emergent agreement field. At 1M, 800 tested overlap pairs have median defect zero and cocycle fraction 1.0, against 0.801 for the shuffled control. These are finite structural receipts. Strict neutral bulk, Einstein-branch entry, physical particles, and physical CMB remain open; the public-data rows above are separately classified paper-side comparisons, not simulator-emitted physical predictions [C7].

3. From closure coordinates to physical structure

The local pixel closure

The SL-3 estimate of the pixel from measured α is

$$P = 1.630968209403959\ldots,$$

a declared input of the claim lattice. The forward model returns its own closure point,

$$P_{\mathrm{fwd}} = 1.630972095858897\ldots,$$

the current forward-model closure point. Their difference, $P_{\mathrm{fwd}} - P \approx 3.89 \times 10^{-6}$, is the loop residual, carried as closure rows CL-1/CL-2 of the closure ledger [C17]. A trial pixel P_k produces a field-width readback $R_P(P_k)$, which the pixel chart turns into the next iterate:

$$\alpha_k = R_P(P_k), \qquad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k\sqrt{\pi}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

On the selected particle branch, the internal readback chain is

$$P \longmapsto \alpha_U(P) \longmapsto A_Z(P) \longmapsto A_T^{\mathrm{src}}(P), \qquad \alpha_{\mathrm{in}}(P) = \frac{1}{A_T^{\mathrm{src}}(P)}.$$

The closure equation the principle set requires to hold exactly (CL-1) identifies the geometric and internal readings; its current solution is P_{fwd}, outside the SL-3 basin:

$$P_{\mathrm{fwd}} = \varphi + \frac{\sqrt{\pi}}{A_T^{\mathrm{src}}(P_{\mathrm{fwd}})},$$

$$\alpha_{\mathrm{root}} = \frac{P_{\mathrm{fwd}} - \varphi}{\sqrt{\pi}} = \frac{1}{A_T^{\mathrm{src}}(P_{\mathrm{fwd}})}.$$

The root is no longer a witness: the interval contraction certificate [C1] proves by direct Banach in outward-rounded interval arithmetic that the declared map is a contraction ($L \leq 0.0724$) on an explicit alpha interval containing exactly one fixed point,

$$\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots,$$

enclosed to width 7.2×10^{-24}, with the SU(2)/SU(3) edge-sum tails bounded by geometric majorants so the enclosure covers the infinite-cutoff sums. The earlier printed tail 136.994835164621... came from an unconverged run and is superseded beyond digit 9 (ledger row CL-6, closed: the printed-pair identity $\alpha_{\mathrm{root}} = (P_{\mathrm{fwd}} - \varphi)/\sqrt{\pi}$ now holds to 35+ digits under converged precision-100 reruns, enforced by a CI test). The certificate is stage 2 of the basin-then-contract protocol; the stage-3 landing verdict is unchanged: the certified fixed point lies outside the SL-3 basin by the missing hadronic transport term (CL-1). The source-audit solve is reproduced by the solver's no-hadron-closure mode in the code release [C1].

The fine-structure audit has four distinct coordinates, each a closure-status reading and none a derivation of α. The certified source root is $136.994835177413\ldots$ (CL-1); the certified self-consistent gauge-width fixed point is $137.035660136946\ldots$ (CL-2, residual 2.5×10^{-6} to measurement); the historically displayed $137.0359595008\ldots$ is a mixed-provenance display packet (inner value at P_{fwd} plus α_U at the SL-3 pixel), not a fixed point of any single declared map, and stays on record as a display packet only; and the measured endpoint is the SL-3 input. An executed empirical closure using external $e^+e^- \to$ hadrons input returns

$$\alpha_{\mathrm{emp}}^{-1}(0) = 136.2636589431, \qquad \alpha_{\mathrm{emp}}^{-1}(0) \in [136.1583832834, 136.3690975240].$$

The measured endpoint is $\alpha^{-1}(0) = 137.035999177(21)$, outside that interval. The same-scheme correction lies in

$$[0.6485541111, 0.8547920666]$$

inverse-alpha units. The declared-empirical compilation behind that payload integrates to $\Delta\alpha_{\mathrm{had}}^{(5)}(M_Z^2) = 0.02676(75)$, within 1.1σ of the KNT19 reference 0.02761(11) [S10,C12]; at the physical on-shell anchor the endpoint residual equals this compilation undershoot, so the interval miss is attributed to the source hadronic spectral backend, which is work in progress. The export's alpha-free spacelike packet is 3.661(103) inverse-alpha units, with requadrature consistency gate 1.5×10^{-7} at tolerance 2×10^{-5} [C12]. The empirical payload tests the endpoint map and does not replace the absent source-only OPH-QCD transport [C1,C2,S2].

The global capacity closure

Let Ω_N^{sc} be the closed-screen normal forms supported at capacity N. The global source map compares their record capacity with the screen budget and applies Minimal Admissible Realization (MAR):

$$S(N) = \log|\Omega_N^{\mathrm{sc}}| - N, \qquad N_\star = \mathrm{MAR}\arg\max_N S(N).$$

The selected endpoint fixes the horizon normalization through

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\mathrm{cell}} = \frac{3\pi P_{\mathrm{fwd}}}{N_\star}, \qquad a_{\mathrm{cell}} = P_{\mathrm{fwd}} \ell_\star^2.$$

Thus the local loop records the candidate pair $(P_{\mathrm{fwd}}, \alpha_{\mathrm{root}})$, while the global loop gives the paired symbolic capacity and horizon readout $(N_\star, \Lambda_\star)$ [O1,O4]. SL-4 admits one capacity N. The electroweak-refined capacity bridge returns $N_{\mathrm{CRC}}^{\mathrm{EW}} = 3.5324 \times 10^{122}$, while the SL-4 estimate from measured Λ at Planck parameters is approximately 3.313×10^{122}: an open 6.6% contradiction, carried as closure row CL-3 of the closure ledger [C17][C3]. The contradiction stands until the EW-bridge/capacity connection is derived or the row is retired; no second capacity enters the claim lattice, and a physical claim must propagate the cosmological posterior and the endpoint-selection provenance.

The compact-gauge branch

The gauge derivation starts with individually transportable seeds selected by zero-obstruction transport. A common strict representative fixes the tensor category carried by those seeds. Surjective pullback functors and forgetful fibers construct its refinement limit, and Doplicher–Roberts/Tannaka reconstruction reads off the compact group. MAR selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel then fix the hypercharge lattice and quotient [O4]. The result is

$$G_{\mathrm{SM}} = \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3.$$

The selected hypercharge lattice is

$$\begin{gathered} Q_i = (3,2)_{1/6}, \qquad u_i^c = (\bar{3},1)_{-2/3}, \qquad d_i^c = (\bar{3},1)_{1/3}, \\ L_i = (1,2)_{-1/2}, \qquad e_i^c = (1,1)_1, \qquad H = (1,2)_{1/2}, \qquad i = 1,2,3. \end{gathered}$$

The Higgs doublet is the Borel–Weil carrier

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2, \qquad \mathrm{Stab}(\langle H \rangle) = \mathrm{U}(1)_Q.$$

Its connected adjoint is

$$(8,1,0) \oplus (1,3,0) \oplus (1,1,0),$$

which fixes charge quantization and excludes mixed $(3,2,\pm 5/6)$ connected adjoint generators. The same branch supplies the electroweak integer

$$\beta_{\mathrm{EW}} = N_c + 1 = 4.$$

The QCD-free hierarchy witness

The source-audit hierarchy witness reuses the pixel candidate and the compact-gauge integers. Its matched unified-width coordinate is

$$\alpha_U(P_{\mathrm{fwd}}) = 0.041124247441816685.$$

The screen count begins with a triangulated S^2. For curvature charge $q_v = 6 - \deg(v)$, Euler's identity and $3F = 2E$ give

$$\sum_v q_v = 6V - 2E = 12.$$

Refinement-stable MaxEnt realizes the twelve units as twelve equivalent fivefold defects, and edge-center completion turns them into twelve central ports. Reversible write/check orientation doubles the gauge-algebra dimension:

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24.$$

With local cell entropy $\ell_{\rm cell} = P_{\rm fwd}/4$ and $\beta_{\rm EW} = 4$, the electroweak share of screen depth is

$$\Gamma_{\rm EW} = \beta_{\rm EW}\ell_{\rm cell}\frac{\log(N/\pi)}{12} = \frac{P_{\rm fwd}}{12}\log(N/\pi).$$

The corresponding transmutation law is

$$\frac{v}{E_\star} = P_{\rm fwd}^{-1/2}\exp\left[-\frac{2\pi}{\beta_{\rm EW}\alpha_U(P_{\rm fwd})}\right] = 2.0198114078576331 \times 10^{-17},$$

evaluated at the certified $P_{\rm fwd}$; the companion value at the SL-3 working pixel is $2.0199803239725553 \times 10^{-17}$. This is the dimensionless source-audit witness. A GeV weak scale additionally requires an independently source-closed physical interpretation and normalization of $E_\star$. The exact parametric law and the 12/24 counts are theorem outputs; the numerical endpoint has the certificate class `source-audit branch witness` [O5,C3]. The weak scale is a settled-form readout with naturality defect

$$\epsilon_H = 0, \qquad \epsilon_H \in [0, 0].$$

At the one capacity of SL-4, the hierarchy relation $\alpha_U \log(N/\pi) = 6\pi/P$ holds iff N equals a P-determined value 6.6% in N units above the Λ-located capacity; the relation is carried as closure rows CL-3/CL-4 of the closure ledger [C17], never as a solved hierarchy.

Conditional W, Z, top, and Higgs masses

The weak-boson pair is read on the strict source-audit branch, with no measured mass entering. The top and Higgs are read on the double-criticality branch: the criticality law $\lambda = 0$, $\beta_\lambda = 0$ at one source scale fixes the top Yukawa from the gauge couplings, so the top is a determined electroweak-sector quantity, not a free flavor parameter, and the Higgs quartic follows on the same curve [O5,C4]:

$$\begin{aligned}(m_W, m_Z)_{\rm OPH} &= (80.330,\ 91.119)\,\text{GeV},\\ (m_t, m_H)_{\rm OPH} &= (172.63,\ 125.77)\,\text{GeV}.\end{aligned}$$

The opening scoreboard carries the dual display for these four endpoints: relative offsets -4.9×10^{-4}, -7.6×10^{-4}, $+2\times10^{-4}$, and $+5.1 \times 10^{-3}$, with pulls of approximately 2.9σ, 35σ, 0.1σ, and 5.8σ in measurement sigma. The m_Z and m_H rows are excluded as predictions at current precision and carried as closure tension CL-5; the top–Higgs lane rests on the D11 census and is labeled calibration. The weak-boson pair is an exact implication of the five quotient-transport premises, conditional on the QT1–QT5 carrier certificate and the discrete two-law choice (ambiguity 9.3 MeV on M_W, 5.1 MeV on M_Z). The top carries the sharpest match in the table, +0.02 percent, with the Higgs companion at +0.5 percent. Both rest on the boundary-scale selection, a theorem modulo two finite carrier facts (CF1/CF2) that are the same D11 census the W/Z law consumes; the three-loop implied scale is the registered discriminating test. None is a promoted physical complex-pole mass.

The determined-top law carries a fit-free relation check. Holding the two-loop criticality curve fixed and inserting the measured top mass returns $m_H = 125.72\,$GeV, +0.47 percent from the measured 125.13 GeV and inside the declared ±1.5 percent matching band, with implied boundary scale 4.75×10^{17} GeV [C4]. The chain behind the branch is short. The pixel record $(P_{\rm fwd}, E_\star)$ fixes the two anchors μ_U and $E_{\rm cell}$ in closed form; the anchor-reconciliation theorem (exactly quadratic record cost on the one-loop chart, capacity-weighted log-mean placement) selects the boundary

$$\mu_b = \sqrt{\mu_U E_{\rm cell}} = E_\star e^{-\pi} P_{\rm fwd}^{-1/6} = 4.86 \times 10^{17}\ \text{GeV};$$

double criticality at μ_b fixes $y_t(\mu_b) = 0.388012$ and $\lambda(\mu_b) = 0$; two-loop running returns $\lambda(m_t) = 0.130064$ and the endpoint pair $(m_t, m_H) = (172.629, 125.771)\,$GeV with declared bands ±2.07 (truncation) and ±1.89 (matching) GeV. A flow-internal alternative is a certified no-go: $d\beta_\lambda/d\ln\mu$ has no root on $[10^{16}, 1.3 \times 10^{19}]\,$GeV (minimum 4.4×10^{-5}), so the boundary scale cannot come from the running itself and has to come from the observer-patch side. The carrier census verifies both finite carrier facts at model level; the surviving gate is the faithfulness export of the pre-repair carrier, and the four-candidate boundary-scale registry is frozen before any three-loop computation exists [C4].

Conditional OPH derivation of the empirical Koide relation

The charged-lepton pole masses empirically satisfy Koide's relation [S3],

$$Q = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{2}{3}.$$

OPH gives a conditional derivation of this value from a finite response packet. Assume a positive Hermitian C_3 square-root-mass carrier,

$$C = aI + \rho(e^{i\delta}R + e^{-i\delta}R^2), \qquad R^3 = I,$$

Fourier diagonalization gives $\sqrt{m_k} = \sqrt{s}[a + 2\rho\cos(\delta + 2\pi k/3)]$ and

$$Q = \frac{1 + 2(\rho/a)^2}{3}.$$

The finite OPH response uses $\mathcal{V}_K = \mathbf{1} \oplus \chi \oplus \bar\chi$, a two-state orientation record, and the connected algebra $B(\mathcal{V}_K \otimes \mathbb{C}^2) \simeq M_6(\mathbb{C})$. A one-state record cannot distinguish an orientation from its reverse, so the two-state record is minimal. Its MaxEnt weights give the singlet action $S_0 = 0$ and the selected charged-orientation action $S_c = \ln 2$. The event

$$E_+ = P_0 \otimes I_2 + P_c \otimes q_+$$

has two rank-two blocks. Born–Lüders conditioning gives $p_0 = p_c = 1/2$. The tracial GNS map $e_0, e_+, e_- \mapsto I, R, R^2$ is unitary, so its canonical square-root amplitude has $a = \sqrt{p_0}$ and $\rho = \sqrt{p_c/2}$. Consequently

$$\frac{\rho}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}.$$

This proves $Q = 2/3$ for the finite source packet. Let $M_F = X_F^2$ be its positive source mass response. At finite cutoff, suppose an accepted physical charged sector supplies three left and three right family modes, positive kinetic metrics, a committed neutral Higgs direction, and faithful trace-preserving u.c.p. checkpoint maps Φ, Ψ with $\Psi\Phi = \mathrm{id}$. If they preserve chirality, the regular C_3 record, the public records, and refinement, reversible-checkpoint rigidity forces $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$. It follows that

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

The square-root response and its block powers are preserved, so the finite source balance gives canonical source-scale $Q = 2/3$. The extended branch $\mathrm{OPH}_{\mathrm{ch}}^+$ adds graded physical completion, quotient source-law selection, and a certified charged QFT self-map. Its dressed readout gives $Q = 2/3$ on the balanced C_3 fixed point and $Q = [1 + e^{-2\chi_\star}]/3$ on the attenuated C_3 branch; an off-plane response uses the full operator. These are additional branch conditions beyond OPH5 [O4,O5,C10]. The phase is independent of Koide. Setting $\delta = 2/9$ as a separate continuation input gives $m_\mu/m_e = 206.770315972729\ldots$ and $m_\tau/m_e = 3477.47283710460\ldots$, differing from the PDG-2026 pole-mass ratios by approximately 9.83 ppm and 30.94 ppm [S8].

Absolute charged-lepton mass intervals

The alpha-transport inversion turns the same family functional into absolute masses. It consumes the measured Thomson endpoint as a declared input together with the frozen one-loop anchor, the certified anchor-gap interval, and the declared-empirical hadronic payload, and inverts for the family scale $\kappa = \ln(m_e/m_e^{\mathrm{wit}})$. The certified result is $\kappa \in [-0.3455, +0.3371]$ with empirical-closure mass intervals

$$m_e \in [0.362, 0.716]\,\mathrm{MeV}, \quad m_\mu \in [74.8, 148.0]\,\mathrm{MeV}, \quad m_\tau \in [1.258, 2.489]\,\mathrm{GeV},$$

each containing the measured mass, with central values (0.5089 MeV, 105.22 MeV, 1.7695 GeV) a common 0.42 percent from measurement. No lepton mass enters the inversion. At the physical on-shell anchor the residual equals the hadronic-compilation undershoot against the KNT19 reference, so the interval width is attributed to the source hadronic backend [C10,C12].

The companion stratum theorem fixes the shape question. On the balanced icosahedral carrier the fivefold-axis and threefold-axis strata carry exactly doubly degenerate quadrupole spectra and the twofold stratum is simple, so three distinct charged-lepton masses force the minimizing orbit off the high-symmetry axes. The stratum classification, the target shape, and the nonempty coefficient locus are closed mathematics; source emission of the coefficient packet is work in progress [C10].

Strong-interaction scale

Dimensional transmutation of the source strong coupling, with no hadronic input, gives $\alpha_s(M_Z) = 0.11834$, $+0.5\sigma$ from the public reference, and fixes the strong scale

$$\Lambda_{\mathrm{QCD}}^{(3)} = 334.8\ \mathrm{MeV},$$

within $\sim 1\%$ of the measured 338(12) MeV, and the published lattice ratio $m_N/\Lambda_{\mathrm{QCD}}$ places the nucleon at 929 MeV, -1.0% from measurement on the external-QCD-conditional branch [C2,C7].

Quark-sector ratios on the conditional register branch

The down-type family is read from the charged-lepton carrier through the register Clebsch triple (1, 1/3, 3) at the unification boundary, with a full one-loop seven-Yukawa system and banded QCD mass running. With no quark input, the intra-sector outputs land at the ten-percent scale: the Gatto–Sartori–Tonin angle is

$$\sqrt{m_d/m_s} = 0.2086 \quad \text{against the measured Cabibbo } 0.2251,$$

a -7.3 percent offset, with $m_s/m_d = 22.97$. The overall normalization overshoots ($m_b(m_b) = 6.03\,\mathrm{GeV}$ against $4.18\,\mathrm{GeV}$); this is the one-loop b–τ tension and is carried openly as the register-selection and threshold-packet gap [C11,S11]. The up-type integer-exponent law family is removed by a frozen negative scan and recorded as such: none of the four frozen bases admits an integer exponent pair for charm and up, so charm and up are research-open [C11].

4. Geometric gravity normalization and the SI readout

On the declared Einstein branch, the edge-entropy/area-law theorem identifies the coefficient of the local Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle$$

and its Newton–Poisson limit with the observer-cell ratio

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}}.$$

The local pixel branch supplies

$$a_{\mathrm{cell}} = P_{\mathrm{fwd}} \ell_\star^2, \qquad \bar{\ell}_{\mathrm{shared}} = \frac{P_{\mathrm{fwd}}}{4}.$$

Substitution gives

$$\frac{G_{\mathrm{geom}}}{\ell_\star^2} = \frac{P_{\mathrm{fwd}}}{4(P_{\mathrm{fwd}}/4)} = 1, \qquad \boxed{G_{\mathrm{geom}} = \ell_\star^2}.$$

The pixel number cancels because it counts both cell area and shared edge entropy. The factor 4 follows from the ratio between the structural Einstein coefficient and the Newton–Poisson convention [O4]. This result is the OPH normalization in geometric units.

Newton coupling and the SI clock readout

A cesium clock translates the geometric identity into SI units:

$$\gamma_\star = \frac{\ell_\star \nu_{\rm Cs}}{c} = \frac{\varepsilon_{\rm Cs}}{2\pi}, \qquad G_{\rm SI} = \frac{c^3 \ell_\star^2}{\hbar} = \frac{c^5 \varepsilon_{\rm Cs}^2}{4\pi^2 \hbar \nu_{\rm Cs}^2}.$$

For the selected display coordinate $\gamma_\star = 4.955974336548419\ldots \times 10^{-34}$, this gives

$$\boxed{G_{\rm SI} = 6.674299995910528 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}}.$$

The 2022 CODATA value is $6.67430(15) \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ [S1]. The agreement is a unit-bookkeeping identity on the selected scale bridge (SL-5): the display coordinate $\gamma_\star$ is chosen on that bridge, so G is an input via the bridge, not an output. Evidence class: exact geometric normalization plus SL-5 bridge input display. The source-promotion certificate is the cesium record stack $\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\rm abs} + \mathcal{R}_{\rm QCD/nuc}^{\rm 133Cs} + \mathcal{R}_{\rm atom}^{\rm 133Cs}$ [O1,O4,C3].

5. Landmark solution map

The table states the positive result first and records its evidence class once. Paper numbers refer to the numbered OPH papers in the References block.

Physics problem	OPH solution on the declared branch	Evidence class	Paper
Observer and measurement problem	An observer is a bounded self-reading record boundary. Compatible local readouts participate in the repair map that returns the public world $T(\mathfrak{U}) = \mathfrak{U}$.	Internal theorem	1,2
Quantum probabilities and update	The finite central record algebra yields the Born rule, Lüders conditioning, and $\lvert S_{\rm CHSH}\rvert \le 2\sqrt{2}$ on the two-wing event surface.	Internal theorem	1,3
Time, Lorentz symmetry, and three space dimensions	Cap modular flow supplies the relative clock; $\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$, and $H^3 \cong \mathrm{SO}^+(3,1)/\mathrm{SO}(3)$ has dimension three over a $3+1$ event base.	Receipt-conditional theorem	4
General relativity	Null translations, local stress, generalized-entropy stationarity, and the uniform small-ball limit compose to $G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab}\rangle$.	Receipt-conditional theorem	4
Gravity from quantum information	The edge-entropy/area law fixes the Einstein and Newton coefficient: $G_{\rm geom} = a_{\rm cell}/(4\bar{\ell}_{\rm shared}) = \ell_\star^2$.	Internal identity	4
Standard Model gauge and matter structure	DR/Tannaka reconstruction plus MAR, anomaly cancellation, and Yukawa invariance give $(\mathrm{SU}(3)\times\mathrm{SU}(2)\times\mathrm{U}(1))/\mathbb{Z}_6$, the hypercharge lattice, $N_c = N_g = 3$, and one Higgs doublet.	Realized-branch theorem	4
Electroweak breaking and charge quantization	The Borel–Weil Higgs carrier $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$ has stabilizer $\mathrm{U}(1)_Q$; the connected adjoint fixes the Standard Model charge lattice.	Realized-branch theorem	4
Electroweak boson, top, and Higgs masses	The strict source-audit branch gives $(m_W, m_Z) = (80.330, 91.119)$ GeV, pulls $\approx 2.9\sigma$ and $\approx 35\sigma$, with no measured mass entering; m_Z is excluded as prediction at current precision and carried as closure tension CL-5. The criticality law determines the top Yukawa from the gauge sector, giving $m_t = 172.63\,\mathrm{GeV}$ $(+0.02\%)$ with the Higgs companion $m_H = 125.77\,\mathrm{GeV}$ $(\approx 5.8\sigma)$; the lane rests on the D11 census and is labeled calibration. No promoted complex-pole mass.	Source-audit + criticality; CL-5 tension	5
Electromagnetism and force carriers	The unbroken $\mathrm{U}(1)_Q$ branch fixes the electromagnetic connection. With the declared quadratic actions, Maxwell has two transverse modes, perturbative Yang–Mills has $2\dim G$, and Einstein gravity has two transverse-traceless modes.	Conditional action theorem	1,4
Proton stability	The selected product-group gauge structure contains no grand-unified gauge bosons connecting quarks to leptons and therefore excludes gauge-mediated proton decay on the realized branch.	Realized-branch corollary	1,4
Weak/UV hierarchy and Higgs naturality	The 12-port screen, 24-slot oriented register, $P_{\rm fwd}$, and $\alpha_U(P_{\rm fwd})$ emit $v/E_\star \sim 2.02 \times 10^{-17}$ with $\epsilon_H = 0$; the hierarchy relation holds iff the capacity equals a P-determined value 6.6% in N units above the Λ-located capacity (closure rows CL-3/CL-4).	Source-audit witness; open closure CL-3/CL-4	5
Fine-structure source scale	The source/root witness, mixed diagnostic, empirical external-data endpoint, and measured endpoint are distinct closure-status coordinates, none a derivation of α (CL-1/CL-2). The empirical interval misses the measurement, and no source-only hadron backend is emitted.	Closure status + empirical audit	5,6
Dark energy / cosmological term	The global record-capacity closure fixes $\Lambda_\star \ell_\star^2 = 3\pi/N_\star$, while vacuum energy lies in the kernel of the local null readout.	Symbolic closure theorem	1,4
Dark-matter phenomenology	Repair-stress bookkeeping gives the galaxy continuation $\nu_{\rm OPH}(x) = [1 - e^{-\lambda_{\rm collar}\sqrt{x}}]^{-1}$ and the deep limit $v^4 = G M_b a_{\rm eff}$. $\nu(x)$ is the McGaugh–Lelli–Schombert empirical fit [S13] evaluated on non-held-out data: the fixed conditional $\mathbb{Z}_6$ branch matches the aggregate RAR scatter at 0.132834 dex on the same data that shaped the function, and its BTFR slope 4 is 1.80σ from the error-aware public-table fit. Beside this, the Cassini quadrupole excludes the universal law at 19.22σ [S15]; the settled-galaxy branch needs a source-derived applicability theorem.	Empirical-fit continuation with audited public-data checks	7
Yang–Mills gap	The transfer/repair theorem gives the conditional identity $\Delta_{\rm YM} = \Delta_{\rm rep}$ under compact-gauge continuum-branch assumptions that contain the Clay-hard content.	Conditional identity	8
Koide relation and charged-lepton ratios	The standalone derivation in Section 3 gives the finite GNS theorem and states its physical attachment assumptions. The 2/9 phase and mass ratios are compare-only.	Conditional physical theorem; compare-only ratios	3
Absolute charged-lepton masses	The alpha-transport inversion emits certified empirical-closure intervals that contain the measured electron, muon, and tau masses, with centrals a common 0.42% from measurement and no lepton-mass input; the interval width is attributed to the open source hadronic backend.	Certified empirical-closure interval	5,6
Three distinct charged-lepton masses	Stratum theorem on the icosahedral carrier: the fivefold and threefold axes carry exactly doubly degenerate quadrupole spectra, so three distinct masses force the minimizing orbit off the high-symmetry strata; coefficient emission is work in progress.	Closed stratum theorem	5
Quark mass ratios and Cabibbo angle	On the register-Clebsch branch conditional on the lepton carrier, the Gatto–Sartori–Tonin angle lands at 0.2086 against 0.2251 with $m_s/m_d = 22.97$ and no quark input; the normalization overshoot is carried openly, and the up-type exponent family is removed by a frozen negative scan.	Conditional sector, sharp ratios	5
String-vacuum selection	Observer-patch consistency defines a finite sieve over critical-string candidates; the Bouchard–Donagi geometry is selected on its declared gate stack.	Conditional selection program	9

6. Standing kill tests

OPH keeps a public falsifiability map with exact failure conditions [C14]. The sharpest standing forks:

Test	Outcome that kills the branch
Gauge-mediated proton decay	Any observation; the realized product-group branch contains no leptoquark gauge boson to mediate it.
Fourth light generation or fourth color	Any discovery; the realized branch fixes $N_g = N_c = 3$.
Charge-lattice outlier	An elementary particle off the derived hypercharge lattice, or a stable fractional-charge color singlet.
Extra light Higgs multiplets or a light superpartner spectrum	Either discovery; the MAR-selected package carries one doublet and no superpartner spectrum.
Universal static dark-response extension	The displayed galaxy equation, applied without a source-derived applicability or screening reduction to the Milky-Way–Sun system, predicts a Cassini quadrupole about 19.22σ from the published summary on the conditional $\mathbb{Z}_6$ branch. That universal extension is excluded; the explicitly scoped old-settled-galaxy branch is conditional.
Three-loop matching packet	The frozen four-candidate boundary-scale registry either converges onto one candidate or the family is eliminated; the registry predates the computation.
Carrier faithfulness export	The pre-repair observer-patch carrier, exported without tuned coefficients, either matches the frozen equation registry or the top–Higgs branch falls.
Source hadronic backend	The emitted spectral measure either moves the empirical endpoint interval onto the measured $\alpha^{-1}(0)$ and keeps the certified lepton intervals, or the pre-attributed deficit account falls.
Coherence-lift null test	The scoped lift-response continuation fixes the canonical susceptibility $\chi_\nu^{\mathrm{can}} = \exp(-P_\star/24)$; a substrate that reaches the declared coherence contrast with null control channels and no force fraction kills that continuation, with the core reconstruction untouched [C14].
Repair-order countermodel	Two accepted repair orders reaching different normal forms under the stated local-diamond and completeness hypotheses kill the consensus mechanism.
Closure uniqueness	A second solution of the P or N closure, or a demonstrated target leak in a declared source map, kills the compression claim. A second P fixed point anywhere on the declared physical domain is arithmetically excluded by the global uniqueness certificate [C1]; the standing fork moves to domains beyond the declared one.

7. Compression and closure status

The evidential standing of the claim lattice is carried by two maintained repository artifacts, cited here in place of any probability estimate.

Compression scorecard [C17]

The metric is $M - K$: independent landed basins M against settings K, counted under the claim rules of the strange-loop principles [C17]. The input side carries the three continuous settings (P from measured α, N from measured Λ, the cesium SI bridge $\gamma_\star$), the basin-locating measurements consumed by current chains, and the discrete structural selections with their menu sizes. The output side currently carries the structural zeros (photon, gluon, graviton mass on the stated action branches) and the hypercharge lattice conditional on its declared clause inputs; no further row qualifies. The current standing is $M - K$ deeply negative. Rows counted as inputs include α, Λ, and G (the SL-3/SL-4/SL-5 estimates), the exact W/Z and Higgs/top lanes (consumed their targets), and the RAR function (fitted to the comparison data).

The closure ledger [C17] records the equations the principle set requires to hold exactly, with residuals in relative units and in measurement sigma: the open pixel-loop term (CL-1/CL-2), the one-capacity contradiction (CL-3/CL-4), the forward electroweak residuals (CL-5), the printed pair identity (CL-6), and the unconstructed capacity readback map (CL-7). Under SL-0 each residual is a measurement of the hypothesis. A row closes only by a blind computation landing inside the stated basin with a contraction certificate, never by relabeling.

Two experiments are decisive. First, the blind Ward-projected hadronic transport against the frozen target in the timestamped target archive [C18]: the completed source chain must supply $\Delta_{\mathrm{source}} = 0.0414658\ldots$ inverse-alpha units at the endpoint without reading the target (CL-1/CL-2). The governing target is the v2 file (precision-160 recomputation, closed-form pass tolerance, payload-work-start rule; v1 and its addendum stay on record beside it), committed and externally timestamped (OpenTimestamps proofs alongside it, 2026-07-14); the comparison happens once, publicly, after the payload is emitted. Second, the forward electroweak emission: the repaired chain from the principles, with preregistered revisions, must move (M_W, M_Z) from $(80.330, 91.119)$ into the measured basins $(80.3692(133), 91.1880(20))$ (CL-5).

Pricing the coincidence

The forward record invites the question: what are the odds that a no-dial architecture lands this pattern by accident? The question admits a disciplined answer. Reference model, declared in full: under the coincidence hypothesis each independent zero-dial output lands log-uniform inside one decade centered on the measured value, so a landing within relative accuracy ε costs $2\varepsilon/\ln 10$. Concessions charged against the theory before multiplying: the W and Z masses count as one event at the looser accuracy; the hierarchy relation is dropped as correlated with the capacity row; every conditional, retrospective, or externally supported row (top, nucleon, tilt, structural zeros, the exact gauge lattice) is excluded outright; the six structural selections on the surviving chains are charged at menu size four each ($4^6 \approx 4{,}100$) although two of them are now consistency-forced; and a reporting look-elsewhere factor of 100 is charged on top. Four events survive this triage: the fine structure constant at 3.0×10^{-4}, the electroweak pair at 7.6×10^{-4}, the record capacity at 6.6×10^{-2}, and the strong scale at 10^{-2}. The product is 8.5×10^{-11}; after the full 4.1×10^5 debit the priced coincidence probability is about 3×10^{-5}: one in thirty thousand, under assumptions selected to favor coincidence at every fork. Reading the electroweak pair as two events and the hierarchy relation as its own event prices it at 7×10^{-9}, one in 140 million. No posterior is printed: a posterior needs a prior over universes, and this document does not pretend to own one. What it owns is the likelihood statement above, with every assumption on the table, and the observation that the structural record (a forced Lorentz group, a forced gauge quotient, three colors, three generations, one Higgs) is priced at zero here despite being the part no random formula generator produces.

The odds question, converted. Beyond the pricing above, the question has been turned into an experiment. Coincidence spreads the blind hadronic completion over its full prior range; SLH commits it, in advance, to a window 2.1×10^{-8} wide in Δ_{source} units (half-width 2.1154×10^{-8}, propagated in closed form from the CODATA α^{-1} uncertainty and printed in the governing v2 target file). One outcome collects evidence no referee can touch; the other ends the formulation. This is the strongest form the odds question admits, and the first-principles bracket already contains the required value.

The 2026-07-14 computations sharpened the program from both sides. The payload harness for the hadronic transport is built to the declared contract, and its first-principles bracket, computed from the chain's internal masses with no measured hadronic input anywhere in the cone, contains the required screening value: first principles currently permit the loop to close. Closing the window itself needs the hadronic moment to 4×10^{-9} relative precision, beyond every current method; the blind-run protocol for the day a method reaches it is written into the payload directory. On the capacity side the coupling theorem is now proven modulo three declared premises, with the tick-projection identity discharged and the conditional fixed point certified; one counting-theorem-shaped premise, the balance condition CP-1, stands between CL-7 and its collapse into the live 6.6% capacity test (CL-3).

The remaining program is finite and enumerated. The seven ledger rows reduce to four generating objects, the hadronic transport (CL-1/CL-2), the capacity readback map F with its contraction certificate (CL-3/CL-4/CL-7), the repaired forward electroweak chain (CL-5), and solver hygiene (CL-6), plus one further object outside the ledger, a source-derived flavor-orbit selector for the mass sector. Every one of the five is a computation or a construction; reinterpretation closes nothing. The distance between the current standing and the establishment criterion is exactly this list.

The assent structure of this document is deliberately mechanical. A reader who grants the six principles and verifies the cited artifacts retains exactly one degree of freedom: the closure ledger. The other exits are closed. The fixed points cannot be relocated (L3), executed failures cannot be reclassified (claim rule 7 [C17]), and no OPH surface may claim more than the ledger and scorecard record. When the ledger is driven to zero by blind completions with uniqueness certificates and the scorecard turns positive, declining the identification of the closure output with our universe requires striking a principle, and each principle names the observation that would justify striking it. Until then, this document claims exactly the current standing and arms the computations that decide.

Where the proof stands (2026-07-15)

Proven: the pixel closure has exactly one solution per readout map on the declared physical domain, by interval arithmetic; the printed-pair identity holds to 35 digits under CI; the SM coefficient triple and $\beta_{\mathrm{EW}} = 4$ are consistency-forced, with the alternatives structurally excluded. Reduced to single named objects: the capacity closure is one coupling theorem (G2-GAP-1) whose truth forces N to the bridge value; the electroweak tension equals a 3×10^{-5} pixel shift and its one-loop menu is exhausted. Consistent and waiting: the first-principles hadronic bracket contains the required screening; closing the window needs 4×10^{-9} relative precision on the hadronic moment, beyond every current method, and the blind protocol for that day is frozen and timestamped. The theory is one theorem and one measurement window away from exact closure, and it cannot be adjusted while it waits.

The establishment criterion is the one stated in the strange-loop principles [C17]: overdetermination and survival: uniqueness certificates for the fixed points, the closure ledger driven to zero under blind completions beginning with the hadronic transport, and a compression scorecard in which independent landed basins outnumber settings. Nothing else establishes SLH, and no wording substitutes for these artifacts.

References

Data and standards

[S1] CODATA Task Group (2024). "2022 CODATA recommended values: Newtonian constant of gravitation." NIST Reference on Constants.

[S2] CODATA Task Group (2024). "2022 CODATA recommended values: inverse fine-structure constant." NIST Reference on Constants.

[S3] Koide, Y. (1982). "Fermion–boson two-body model of quarks and leptons and Cabibbo mixing." Lett. Nuovo Cimento 34, 201–205.

[S8] Particle Data Group (2026). "Leptons." Review of Particle Physics summary table.

[S9] NuFIT collaboration (2026). "NuFIT 6.1: three-flavour global analysis of neutrino oscillation data." www.nu-fit.org.

[S10] Keshavarzi, A., D. Nomura, and T. Teubner (2020). "$g-2$ of charged leptons, $\alpha(M_Z^2)$, and an update of the hadronic contributions." Phys. Rev. D 101, 014029.

[S11] Particle Data Group (2026). "Quarks." Review of Particle Physics summary table.

[S12] Planck Collaboration (2020). "Planck 2018 results. VI. Cosmological parameters." Astron. Astrophys. 641, A6.

[S13] McGaugh, S. S., F. Lelli, and J. M. Schombert (2016). "The Radial Acceleration Relation in Rotationally Supported Galaxies." Phys. Rev. Lett. 117, 201101.

[S14] Lelli, F., S. S. McGaugh, and J. M. Schombert (2019). "The Small Scatter of the Baryonic Tully–Fisher Relation." Mon. Not. R. Astron. Soc. 484, 3267–3278.

[S15] Park, R. S., et al. (2026). "Cassini spacecraft navigation data constrain modified gravity in the Solar System." arXiv:2602.17884.

OPH papers All OPH series papers below use release `r1540`.

[1/O1] Mueller et al. (2026). "Observers Are All You Need."

[2/O2] Müller et al. (2026). "Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics."

[3/O3] Müller et al. (2026). "Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH."

[4/O4] Müller et al. (2026). "Recovering Relativity and the Standard Model from Observer Overlap Consistency."

[5/O5] Müller et al. (2026). "Deriving the Particle Zoo from Observer Consistency."

[6/O6] Müller, B. (2026). "The Fine-Structure Constant as an OPH Pixel Fixed Point." OPH technical note.

[7/O7] Mueller, B., and D. Matscheko (2026). "Observer-Patch Holography and the Dark Matter Phenomenon."

[8/O8] Müller, B. (2026). "Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics." OPH technical note.

[9/O9] Müller, B. (2026). "Observer-Patch Holography as a String-Vacuum Selector." OPH technical note.

Executable artifacts

[C1] FloatingPragma (2026). Pixel-closure solver and receipts.

[C2] FloatingPragma (2026). Particle-branch source and audit suite.

[C3] FloatingPragma (2026). Hierarchy theorem package and receipts.

[C4] FloatingPragma (2026). Electroweak calibration receipts.

[C5] FloatingPragma (2026). Consensus schedule benchmarks.

[C6] FloatingPragma (2026). Lean 4/Mathlib theorem artifact.

[C7] Mueller, B. (2026). OPH-FPE simulation, 64k/128k/1M earned run receipts, and the public-data comparison ledger.

[C8] FloatingPragma (2026). Lorentz–Einstein geometry receipts.

[C9] FloatingPragma (2026). SPARC galaxy scorecard and receipts.

[C10] FloatingPragma (2026). Charged-lepton interval and orbit-geometry lanes.

[C11] FloatingPragma (2026). Flavor register lanes and negative scans.

[C12] FloatingPragma (2026). Declared-empirical hadronic spectral lanes.

[C13] FloatingPragma (2026). Theorem gap register.

[C14] FloatingPragma (2026). Falsifiability map and kill conditions.

[C15] FloatingPragma (2026). The OPH consistency stack: selection chain, uniqueness lemmas, generator table.

[C16] FloatingPragma (2026). Capacity readback map F: formal specification, coupling theorem, and certificate schema.

[C17] FloatingPragma (2026). OPH claim registry: strange-loop principles, closure ledger, compression scorecard, consistency stack, proof spine.

[C18] FloatingPragma (2026). Frozen blind targets with external timestamp proofs.